Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)		T = 40 ms **or** f = 25 Hz **or** by implication with correct use of $v = \frac{\lambda}{T}$ **or** $v = f\lambda$, ignoring slips of 10^n (1) v = 0.50 m s^{-1} **unit mark** No ecf (1)		2		2	2	
		(ii)	I.	Same as printed graph. Accept poor shape.		1		1	1	
			II.	Same as printed graph but amplitude 2 mm. Accept poor shape. No ecf.		1		1	1	
		(iii)		$S_1Q - S_2Q = \frac{\lambda}{2}$ or equivalent (1) Waves (arrive) in antiphase / <u>exactly</u> out of phase (1) Displacements [due to S_1 and S_2] cancel [or partly cancel] **or** there is destructive interference, [so she is correct] (1)			3	3		
	(b)	(i)		Substitution and rearrangement: $\sin\theta = \frac{3\times590}{2000}$ accepting slips in powers of 10 (1) $\theta = 62[.25\ ^{\circ}]$ (1)		2		2	2	
		(ii)		3λ **or** 1 770 nm Accept 1.77×10^{-6} m or 1.8×10^{-6} m		1		1		
				Question 5 total	**0**	**7**	**3**	**10**	**6**	**0**